An entry–exit law for dead-neuron resurrection: the smoothed-ReLU limit of training dynamics at an activation boundary, and its confirmation in a small MLP

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Abstract

A neuron "dies" when its pre-activation margin crosses zero on all data; for a hard ReLU the dead unit has exactly zero gradient and never returns. We ask what governs death and resurrection in the smoothed activations used in practice. For shallow softplus networks under gradient flow we prove that, in the sharp limit $\beta \to \infty$ of the activation stiffness, death and resurrection are governed by a hybrid system *with memory*: the dead unit's parameters freeze and reactivation occurs exactly when the signed time integral of the excluded residual returns to zero — the classical entry–exit function — with a universal first-order release coefficient; all constants are verified on one two-neuron instance. The two limit orders do not commute: the discrete ReLU flow freezes dead units forever, the smoothed flow revives them by the area law. We then track the law upward through a pre-registered experimental ladder: a reduced stochastic-schedule model is falsified; two instrument designs are recorded as negatives (a calibrated null fires); and a confirmatory MLP run (400 runs, $T = 0$) reproduces resurrection at $p = 0.50$ with the residual area opening before 13/16 returns, hard ReLU giving zero, and a monotone channel staircase in β. A transformer confirmatory run is in progress.

1 Introduction

Write $\Phi : \mathcal{W} \to \mathcal{F}$ for the parametrization of a network and $J = D\Phi$ for its Jacobian. A neuron whose pre-activation margin is negative on every datum contributes an identically zero column to J: it sits on a stratum of the singular locus where J drops rank. Under a hard ReLU the gradient of the loss in the directions of that unit is exactly zero, so the stratum is *absorbing* — a dead unit stays dead. This is the mechanism behind the empirically documented dormant-neuron and loss-of-plasticity phenomena [**?**, **?**, **?**]. The question of this paper is what replaces the absorbing stratum when the activation is *smoothed*, as it is in every softplus/GELU network actually trained.

Our answer is an exact gating law and its numerical arc. The contributions are:

- **A theorem (§??, Thm. ??).** In the sharp limit $\beta \to \infty$ of a shallow softplus network under gradient flow, the death/resurrection of a neuron is the entry–exit problem of a slow–fast system: the dead unit freezes and reactivates when $\int R\,dt = 0$, where R is the residual excluding that unit. The uniform rate is $O(\log \beta/\beta)$, the release time carries a *universal* first-order coefficient $-1/(cR(t_{\text{exit}}))$, and the law is independent of the smoothing within the monotone-sigmoid class (softplus, erf); GELU is excluded because its derivative is non-monotone. Six lemmas, each with an executable verification, are summarized in §**??** and stated in Appendix **??**; Figs. **??**–**??** show the mechanism and the predicted constants.

- **Non-commutation (§??).** The two limit orders (smooth-then-sharp vs. the pointwise generalized-gradient flow) do not commute. The field-first order is the frozen branch of Eberle, Jentzen, Riekert and Weiss [**?**]; our flow-first limit resurrects. This is the exact mathematical content of "dead ReLU units stay dead, smooth units come back."
- **A pre-registered experimental ladder (§??).** Organized by prediction, with the null controls and the negatives shown: a reduced deterministic-schedule model *falsified* under SGLD (§**??**); two instrument designs recorded as *negatives* because a calibrated null fired (§**??**); the mechanism first observed in an MLP pressure pilot (§**??**); and a *confirmatory* MLP run — 400 runs, area law reproduced with confidence intervals, hard ReLU as the zero-gradient control, monotone staircase in β (§**??**, Fig. **??**). A transformer confirmatory run is in progress (§**??**); its scoping already exposed two new mechanisms (a GELU sign-flip barrier and an upstream channel).

The paper is deliberately honest about scope: the theorem is one two-neuron instance; the confirmatory is one small architecture at temperature zero; the stochastic gate is open and, where we probed it, it *breaks* the deterministic schedule (§**??**).

2 Setup and notation

Data $(x_i, y_i)_{i=1}^n$ in general position; shallow network $f_\theta(x) = \sum_k g_\beta(w_k x + b_k)$ with softplus $g_\beta(z) = \beta^{-1}\log(1+e^{\beta z})$ (general monotone sigmoids in §**??**); quadratic loss $L = \frac{1}{2n}\sum_i (f_\theta(x_i) - y_i)^2$; gradient flow $\dot\theta = -\nabla L$. The parameter β is the *activation stiffness*: as $\beta \to \infty$, $g_\beta \to$ ReLU and $g'_\beta \to \mathbf{1}_{\{z>0\}}$.

Fix one neuron A and one datum j. Its *margin* is $z = \theta_A \cdot (x_j, 1)$; write $c = \|(x_j,1)\|^2/n$ and let

$$R(t) = y_j - f_{-A}\big(x_j; \theta_B(t)\big)$$

be the *residual excluding* A (θ_B collects the other parameters). The *hybrid reference* θ^{hyb} runs the full ReLU flow until the first time t_{hit} with $z = 0$, then freezes θ_A and runs θ_B alone; set R^{hyb} accordingly and $A^{\mathrm{hyb}}(t) = \int_{t_{\mathrm{hit}}}^t R^{\mathrm{hyb}}$ for its signed area. The death of A is exactly the rank drop of J on the stratum $\{z \le 0\}$; the question is the fate of the trajectory on that stratum.

Hypotheses. **(H1)** $R^{\mathrm{hyb}}(t_{\mathrm{hit}}) < 0$ (transversal death). **(H2)** A^{hyb} has a first zero at $t_{\mathrm{exit}} < T$ with $R^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$ and $A^{\mathrm{hyb}} < 0$ on $(t_{\mathrm{hit}}, t_{\mathrm{exit}})$. **(H3)** the other margins of A stay $\le -\delta_1 < 0$ on the episode. **(H4)** no other switching event occurs in $[0, T]$.

3 The entry–exit theorem

Definition 1 (hybrid limit)**.** *θ^∞ equals: the ReLU flow on $[0, t_{\mathrm{hit}}]$; on $[t_{\mathrm{hit}}, t_{\mathrm{exit}}]$, $\theta_A \equiv \theta_A(t_{\mathrm{hit}})$ (frozen) with θ_B the B-only flow; on $[t_{\mathrm{exit}}, T]$, the ReLU flow restarted at $(\theta_A(t_{\mathrm{hit}}), \theta_B^{\mathrm{hyb}}(t_{\mathrm{exit}}))$.*

Theorem 1 ($T2'$-min)**.** *Under (H1)–(H4), for every $\delta > 0$,*

$$\sup_{t \in [0,T]\setminus(B_\delta(t_{\mathrm{hit}}) \cup B_\delta(t_{\mathrm{exit}}))} \|\theta^\beta(t) - \theta^\infty(t)\| = O_\delta\Big(\tfrac{\log\beta}{\beta}\Big), \qquad t^\beta_{\mathrm{release}} = t_{\mathrm{exit}} - \frac{1}{c\,R^{\mathrm{hyb}}(t_{\mathrm{exit}})}\cdot\frac{1}{\beta} + o(\beta^{-1}),$$

where $t^\beta_{\mathrm{release}}$ is the first return of the margin z to 0 after death. The $\log\beta$ in the sup is sharp and is carried solely by the normal coordinate of θ_A inside the episode (the sticking depth $z \sim -\beta^{-1}\log\beta$);

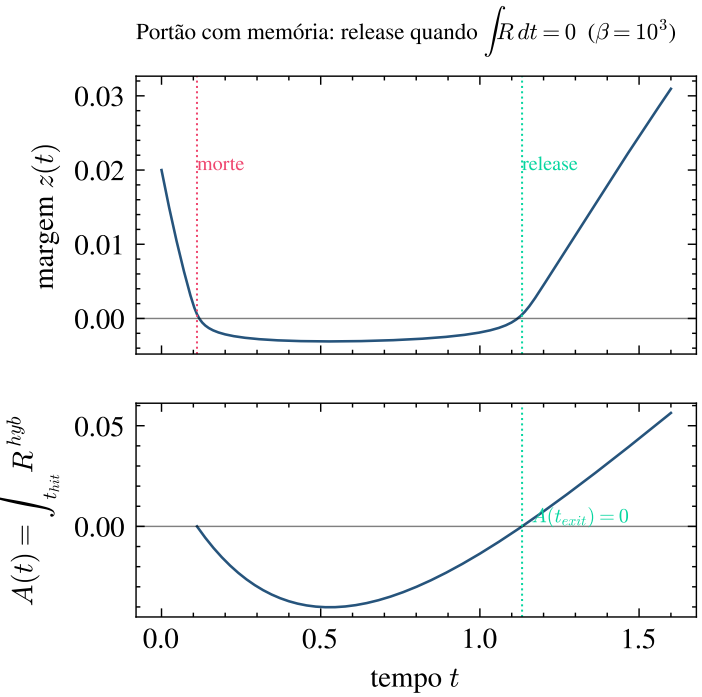


Figure 1: The memory gate at $\beta = 10^3$ (coupled system, reference instance). Top: the margin $z(t)$ dies at t_{hit}, hugs the sticking layer of depth $\sim \beta^{-1}\log\beta$, and releases. Bottom: the hybrid area $A(t) = \int_{t_{\mathrm{hit}}}^{t} R^{\mathrm{hyb}}$; release coincides with $A = 0$ up to $O(\beta^{-1})$ (Thm. **??**). Deterministic Euler integration ($dt = 10^{-5}$); uncertainty is discretization only. Script `t2-prova/figuras_draft.py` (committed).

θ_B obeys $O(\log\beta/\beta^2)$ and the tangential components of θ_A are exponentially frozen. The first-order release coefficient is universal*: in the product (entry delay × area cut) the arrival speed cancels, leaving the reciprocal exit speed — each corner contributes the reciprocal of its transversal speed. The remainder is $o(\beta^{-1})$ as proved. The limit is that of the regularized flow; discrete ReLU gradient descent freezes the dead neuron forever (the limits do not commute, §**??**).*

Proof outline. The episode reduces to a scalar entry–exit problem through six lemmas (Appendix **??**); each has a committed executable symbolic/numeric verification (`passo0--6`, `prova_L1--L6`). In the gauge $w = e^{\beta z}$, fast time $\tau = \beta t$, the margin obeys the quadratic normal form of De Maesschalck–Schecter [**?**] with $\varepsilon = \beta^{-1}$ (Lem. **??**); with $u = 1/w$ and $\Phi(u) = \log u + u$ one has the exact additive quadrature $d\Phi/dt = -c\beta R + c\log(1+w)$, so the layer is held by the *area*, not by the sign of R (Lem. **??**): $u(t) = c\beta|A(t)|(1+o(1))$ returns to $O(1)$ at $t_{\mathrm{exit}} + O(\beta^{-1})$. The B-subsystem is affine and, conditionally on the layer being maintained, tracks the hybrid at $O(\log\beta/\beta^2)$ (Lem. **??**). A coupled continuation (nonempty–open–closed) closes the feedback loop and yields the two-term release law (Lem. **??**); the entry and exit corners contribute mirrored $1/\beta$ terms whose arrival-speed factor cancels (Lems. **??**, **??**). The dead face carries an identically zero field on one side — a degeneracy outside generic piecewise-smooth exit theory [**?**, **?**] and of the exponentially flat type of Kristiansen [**?**]; the canonical Sotomayor–Teixeira regularization [**?**, **?**] requires both one-sided normal components nonzero and so does not cover it. The coupling of the

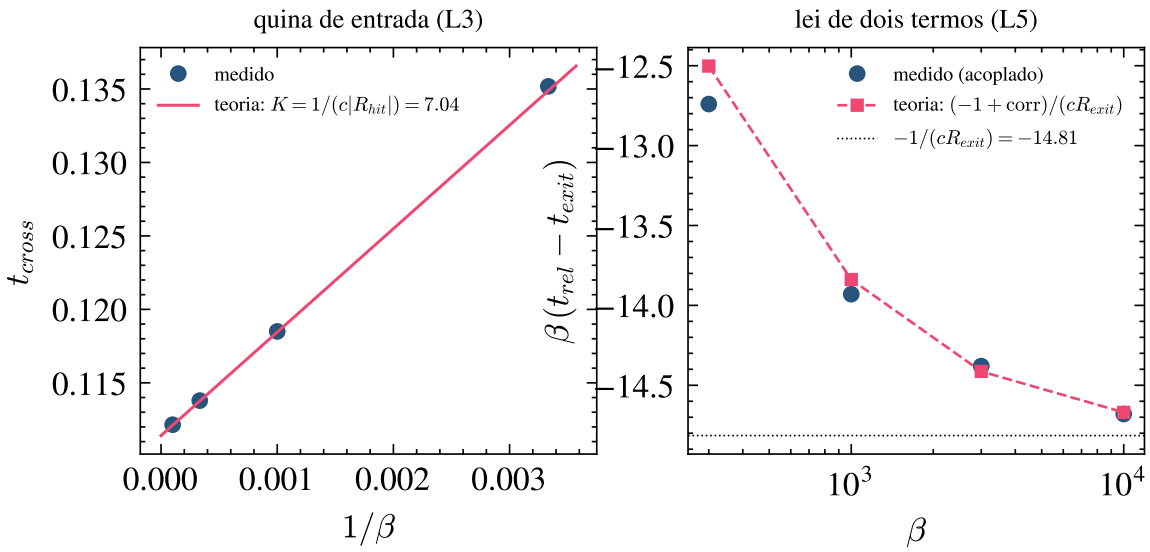


Figure 2: Predicted constants vs. measurements (deterministic runs; uncertainty is discretization). Left: entry-corner crossing times against $1/\beta$; the line is the *proved* slope $K = 1/(c|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|) = 7.04$ (Lem. **??**), not a fit. Right: the two-term release law (Lem. **??**, squares) against the coupled measurements (circles) over $\beta \in [3 \cdot 10^2, 10^4]$; dotted line: the universal first-order coefficient $-1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})) = -14.81$. Script `t2-prova/figuras_draft.py`.

forcing to the flow (the residual is generated by the very trajectory it gates) has, to our knowledge, no analogue in the entry–exit literature [**?**].

Numerical verification (one instance). $X = (0.5, 1.5)$, $Y = (0.5, 1.0)$, $\theta_A(0) = (-0.5, 0.27)$, $\theta_B(0) = (1.144, 0.261)$; $t_{\mathrm{hit}} = 0.1114$, $t_{\mathrm{exit}} = 1.1325$, $\delta_1 = 0.508$. Every empirically measured constant is predicted by the proofs: entry coefficient $K_{\mathrm{teo}} = 7.04$ vs 7.13 fitted (1.2%); the coupled two-term law $(-1 + \mathrm{corr})/(cR_{\mathrm{exit}})$ reproduces the fine-grid coupled sweep $-12.74/-13.93/-14.38/-14.68$ at 1.9%/0.7%/0.2%/0.1% for $\beta = 3 \cdot 10^2 \ldots 10^4$ (universal coefficient $1/(cR_{\mathrm{exit}}) = 14.815$; against -14.68 the leading coefficient alone is at 0.9%); tangential drift at machine zero. All numerics are this single two-neuron instance; breadth is future work.

Regularization independence. The release criterion depends only on $\int R\,dt = 0$, not on the sigmoid: for the erf class ($g'_\beta = \Phi_{\mathrm{gauss}}(\beta z)$) the coupled system releases at the same t_{exit} with $O(\beta^{-1})$ gap (verified). GELU itself, $z\,\Phi_{\mathrm{gauss}}(z)$, has a non-monotone derivative and is *not* a member of the family for any β; §**??** shows this is not a technicality but a distinct mechanism.

4 Non-commutation and the plasticity bridge

The theorem is the flow-first limit: take $\beta \to \infty$ of trajectories. The *field-first* order — flow the pointwise-limit generalized gradient — instead freezes degenerate neurons forever; this is the rigorous branch of Eberle, Jentzen, Riekert and Weiss [**?**], whose selected solution has a non-decreasing degenerate set (the convention $\mathrm{ReLU}'(0) = 0$ and the no-revival mechanism are the same, though their degenerate stratum is the fully-zero one, not literally our functionally dead

neuron). The two orders do not commute: field-first freezes, flow-first resurrects by the area law.

In machine-learning terms this non-commutation is the statement behind an empirical trend: discrete ReLU units that die stay dead — the dormant-neuron and loss-of-plasticity phenomena documented across deep and reinforcement learning [**?**, **?**, **?**], mitigated in practice by recycling dead units or by migrating to smooth activations — whereas the smoothed flow resurrects them by the area law. The channel that reopens a smoothed unit is $e^{\beta z}$, which closes as $\beta \to \infty$; §**??** measures the resulting staircase directly, and the most alive arm ($\beta = 3$) is the smoothness of the GELU family.

5 Experiments

The claims are tested by prediction, each with its null control, and the failed designs are shown rather than hidden. Every confirmatory analysis was frozen by a pre-registration commit before the run; hypotheses that fell were archived, not rewritten. Commits: piloto-pressure `b48b527`; rate pilot `995a87c`; 4a.3 pre-registration `a8f8f8c`; 4a.3 confirmatory `5a961cb`; transformer pre-registration `1b45498`.

5.1 The deterministic schedule is falsified under SGLD

Before any confirmatory claim, we tested whether the deterministic entry–exit schedule survives noise. A reduced "race" model (pre-registered, commit `190d94a`, two adversarial-review rounds) predicts the capture (resurrection) probability of a dead unit from the mean residual schedule; the falsifying run (2700 runs) rejected it. The predicted-zero window was violated decisively: 139 late captures occurred where the model gives exactly zero, and the bin $[0, 0.1)$ saw 150 captures against 1.4 predicted ($z = +131$). Crucially the model is well calibrated where it says capture is *likely* (the $(0.5, 0.9)$ bin, $z = -0.26$); the entire error is on the "impossible" side. The theorem-part of the model held: the observed censoring (2–7%) matched the Brownian-cone prediction in all nine cells — so the failure is of the *hypothesis* (a deterministic schedule), not of the instrument.

An erratum records the next step honestly: the natural fix (a stochastic Ornstein–Uhlenbeck schedule, no free parameter) predicts p(capture) nearly identical to the deterministic one (differences ≤ 0.013) and a late-capture rate of 0.1–1.5% against the $\sim 13.8\%$ observed — insufficient by an order of magnitude. That model was therefore refuted at the predictor stage and *not run*; the leading candidate became the back-reaction of the reactivating unit on the shared residual. The stochastic gate remains open and is out of scope for the theorem, whose regime is deterministic gradient flow.

5.2 Two instrument designs recorded as negatives

The first attempt to see the law in a real MLP failed on the *instrument*, and the failure was caught by a pre-registered null (etapa-4a, commit `6411ac8`, 600 runs, MLP $m \in \{16, 64, 256\}$, SGLD $\eta = 3 \cdot 10^{-4}$). The pure-diffusion arm passed the calibrated test ($m = 16$, $p = 0.002$); by the pre-registered rule this means the ruler is biased and *no* scientific verdict on the substantive predictions may be claimed — the run is **invalid**. Diagnosis (Fig. **??**): the ruler was measuring boundary *chatter*, not the theorem's physical death. 59% of episodes lasted ≤ 5 steps; hard ReLU "resurrected" at the same rate and with the same duration distribution as softplus at $T > 0$ — impossible by gradient (a dead ReLU unit has exactly zero gradient), hence noise threshold-crossings. The real physics lives at $T \to 0$: the only genuinely long episodes (median

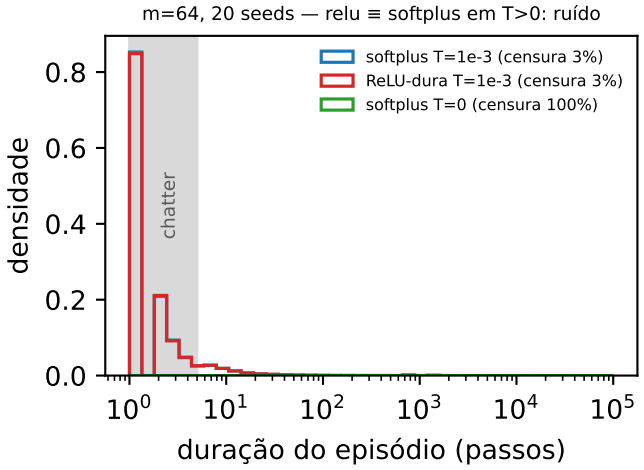


Figure 3: Why etapa-4a is an instrument negative (post-hoc diagnostic, $m = 64$, 20 seeds). Episode-duration densities: at $T > 0$, hard ReLU $\equiv$ softplus (median ~ 4 steps, shaded chatter band) — a signature of noise, not of the zero-gradient theorem; the real physics is the $T = 0$ arm ($\sim 10^3$ steps, 99.6% censored). Script `etapa4-mlp-populacao/figuras_diagnostico.py`.

~ 1110 steps) sit in the $T = 0$ arm, 99.6% censored at the 10^5-step horizon. The calibrated null held false-positive rate near α in i.i.d. worlds, but the real world has per-neuron heterogeneity (non-i.i.d.).

A redesign (etapa-4a.2) was itself rejected at the design stage by the same process (commit `a413b19`): at $T = 10^{-5}$ the ReLU/softplus ratio stayed ≈ 0.95 after debounce (gradient starvation at $\beta = 300$ masked the effect), and the pooled null broke α by 30–92$\times$ in heterogeneous/autocorrelated worlds. The inheritance was explicit: a hysteresis/dwell ruler, a per-neuron stratified null, and a move to the deterministic regime $T = 0$ with a much longer horizon. These negatives falsify neither the area law nor the toy calibration; they falsify two rulers, and they are what forced the working design.

5.3 The mechanism, observed

Moving to $T = 0$ and a sub-parametrized regime (pressure: a plateau ~ 0.15 on the rich target $y = \sin(2\pi x) + 0.3\sin(5\pi x)$, $n = 16$, $\eta = 10^{-3}$, $H = 2 \cdot 10^5$; 64 runs, commit `b48b527`, exploratory) made the mechanism visible. Hard ReLU gave a *structural zero*: 0 resurrections out of 25 deaths in 16 seeds, the frozen (grad $\equiv 0$) branch exact in a real network at $T = 0$. Softplus resurrected by mechanism: at $\beta = 10$ a unit died at step 781 and returned at 139,179; at $\beta = 300$ one died at 3315 and returned at 152,977 (it had parked shallow enough that the $e^{\beta z}$ channel did not vanish). Both dynamical resurrections showed the ordered triple $t_{\text{death}} < t_{\text{area}} < t_{\text{res}}$ (area law 2/2). The gate→return gap is the finite-β rise time ($\sim |z_{\text{park}}|/\text{drift}$), not the asymptotic $O(1/\beta)$ correction. The bottleneck was quantified as the event rate (~ 0.1/run) — which set the entire sizing of the confirmatory.

arm	dyn. deaths	λ/run [boot]	returns	p(ret | death) CP	ordered/returns
sp-β3	35	0.350 [0.220,0.500]	16	0.457 [0.288,0.634]	13/16
sp-β10	27	0.270 [0.160,0.390]	10	0.370 [0.194,0.576]	6/10
sp-β300	19	0.190 [0.100,0.290]	4	0.211 [0.061,0.456]	3/4
ReLU	17	0.170 [0.090,0.260]	0	0.000 [0.000,0.195]	0/0

Table 1: Confirmatory 4a.3 (100 runs/arm, $T = 0$). ReLU gives a structural zero (validity control: any ground-truth return here would invalidate the run). Clopper–Pearson 95% intervals; bootstrap clustered by seed.

5.4 Confirmatory: the area law in a real MLP

The confirmatory (etapa-4a.3) was pre-registered at commit `a8f8f8c` after two more adversarial rounds, with every design number derived programmatically from a 240-run rate pilot (commit `995a87c`); the run is commit `5a961cb`. Grade: {sp-β3 (primary), sp-β10, sp-β300, ReLU} $\times$ $m \in \{4, 6\} \times 50$ virgin seeds (1900–1949) = 400 runs; $T = 0$, $H = 5 \cdot 10^5$; eligible window $[0, 307{,}536]$ ($H - \tau$, τ the $q_{0.9}$ return time). The instrument is a single ground-truth clock with bit-exact parity to the pilot motor; the area of the leader-datum residual is accumulated on that clock, so temporal ordering is structural and the only empirical failure mode of the area test is a *return without area*.

The pre-registered verdicts (Table **??**, Fig. **??**):

- **C1 (existence with rate, primary sp-β3): reproduces.** $N_{\text{elig}} = 30$ ($n^* = 10$), 15 eligible returns, p(ret | eligible) = 0.500 $[0.313, 0.687]$, well above the kill threshold 0.076.
- **C2 (the area law, the heart, sp-β3): sustains.** The leader-drive area opens before the return in 13/16 dynamical resurrections (lower Clopper–Pearson bound $0.544 > 0.5$, ≥ 6 returns). The 3 returns-without-area are real and counted against the law (returns driven off the leader datum — an object of study, not hidden).
- **C4 (channel staircase, estimation, no kill): monotone in β.** p(ret | death) falls $0.457 > 0.370 > 0.211 > 0.000$ from $\beta = 3$ to hard ReLU, with intervals — the activation smoothness controls the return channel, and $\beta = 3$ (the smoothness of the GELU family) is the most alive. This is the direct bridge to the plasticity practice of §**??**.
- **Validity: clean.** ReLU 0/17 dynamical returns — the non-commutation of Thm. **??** measured with $N = 100$.

The honest limits of the confirmatory: $T = 0$ (deterministic), one small architecture, a 1D rich target, $m \in \{4, 6\}$. The stochastic regime is a paper of its own (§**??**). C1's kill is intentionally near-unreachable at this N (a consequence of the frozen $\frac{1}{2}\times$ pilot-lower-bound rule), so in practice C2 is the deciding test; this was signalled in the pre-registration, not discovered after.

5.5 Transformer: two new mechanisms, confirmatory in progress

The same question in a one-layer transformer (FFN over an attended token) does *not* reduce to the MLP, and the scoping (exploratory, commit `d4e4c35`) exposed why, with two facts that reshape the confirmatory:

- **S1 — a GELU sign-flip barrier.** GELU$'(z)$ changes sign at $z^* = -0.7518$ (verified analytically): for $z < z^*$ it is negative, so a deep-dead unit is pushed *away* from the return

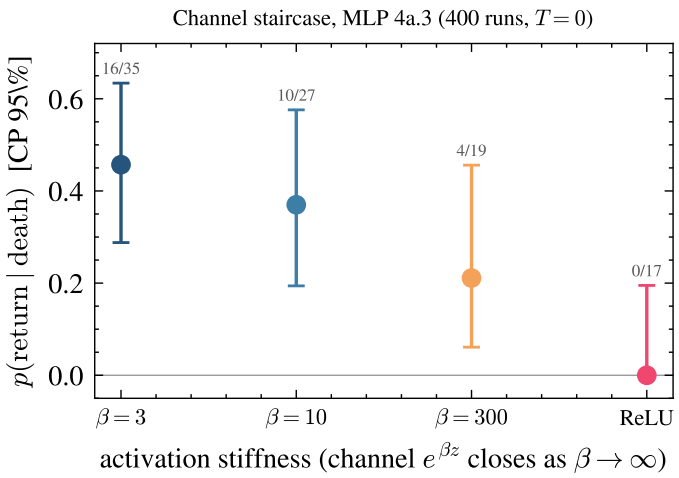


Figure 4: The channel staircase (C4, confirmatory 4a.3, 400 runs, $T = 0$). $p(\text{return} \mid \text{death})$ is monotone in the activation stiffness and reaches a structural zero at hard ReLU (grad $\equiv 0$); the smooth channel $e^{\beta z}$ closes as $\beta \to \infty$. Fractions above each point are returns/deaths; error bars are Clopper–Pearson 95%. Script `figuras_paper.py`.

by its own gradient. Self-resurrection is possible only in the shallow band $(-0.752, 0)$. The MLP's blind area port (which omits g') gets the *sign* wrong on 12/12 GELU episodes; the correct instrument uses the slope-weighted area A_{drive}. GELU is therefore genuinely outside the monotone-sigmoid class of Thm. **??**, not a technicality.

- **S2 — an upstream channel.** The FFN input $h = e + \text{attn}(e)$ is non-stationary, so a hard-ReLU unit with own-gradient identically zero can still revive because h moves. With attention frozen the control is exact: Δmargin $= 0.000$, own_move $= 0$; with attention training, own_move $= 0$ still but the margin climbs $+0.69$ on average (up to $+1.32$). The MLP theorem "dead ReLU = terminal" holds only when the input is fixed; the transformer adds a channel the MLP does not have.

A rate pilot (160 runs, commit `2dba27d`) then fixed the design: natural FFN death is ≈ 0 (units in $d = 16$ input do not die on their own), so the confirmatory is necessarily injection-conditional; the non-stationary channel chatters, forcing a hysteresis instrument ($\delta_h \approx 0.0476$, calibrated from data); the return time is slow ($\tau \approx$ 60–66k), forcing $H = 2 \cdot 10^5$. The upstream channel was already clean in the pilot (Δ = train$-$frozen $= +6/+6/+6/+8$ returns; ReLU $0 \to 8$ with own_move $= 0$).

The transformer confirmatory (etapa-5, pre-registered commit `1b45498`; 800 runs, {sp-β3, sp-β10, GELU, ReLU} $\times$ {attn-train, attn-frozen}$\times$ injection $\{-0.3, -2.0\} \times 50$ seeds 2000–2049; $H = 2 \cdot 10^5$, $T = 0$) **is running at the time of writing**; the following five estimands and their results are pending.

- **[E5-PENDING: E-A validity]** — any sustained return in relu-frozen invalidates the run (theorem: own-gradient $\equiv 0$ and fixed input $\Rightarrow$ terminal). Pilot: 0/40.
- **[E5-PENDING: E-B upstream channel]** — $\Delta = p_{\text{sust}}(\text{train}) - p_{\text{sust}}(\text{frozen})$ per arm; reproduces if the bootstrap lower bound > 0. Stricter under hysteresis than the pilot; a genuine test, not a stamp.

- **[E5-PENDING: E-C transfer of the theorem class]** — sp-β3 frozen, shallow injection: p(sustained return | inj); transfers if $\geq n^* = 6$ sustained returns. This is the direct analogue of the MLP C2.
- **[E5-PENDING: E-D GELU barrier]** — gelu-frozen, deep (-2.0) vs. shallow (-0.3); the barrier predicts shallow $\gg$ deep in the own channel, with a depth-matched control against sp-β3. The pilot at -1.0 was a tie (did not decide).
- **[E5-PENDING: E-E β staircase]** — p(sustained return) across $\beta \in \{3, 10\}$, frozen and train (estimation, no kill).

The scoping verdict is itself a result: the MLP mechanism does not transfer directly to the GELU FFN; the sign-flip barrier obstructs self-resurrection and the upstream channel is a new route. The confirmatory tests the theorem class (softplus-β) on the FFN and treats the GELU barrier and the upstream channel as estimands of their own.

6 Limitations

We state them explicitly. **(i)** The theorem is proved for one two-neuron instance; every constant is verified there and breadth is future work. **(ii)** The regime is $T = 0$ deterministic gradient flow throughout the theorem and both confirmatories; the stochastic (SGLD) gate is open and, where probed, *falsifies* the deterministic schedule (§**??**) — late captures recur past the deterministic window at a rate the reduced stochastic model under-predicts, leaving the reactivating unit's back-reaction as the leading candidate driver. **(iii)** Scale: the confirmatories are small networks ($m \leq 6$ MLP units, $m_{\text{ffn}} = 4$ FFN units) with rich 1D targets, chosen so that a single unit can be tracked with a ground-truth clock. **(iv)** GELU is outside the monotone class; the transformer confirmatory is in progress and its five estimands are pending. **(v)** The cone apex (corank ≥ 2), where weight decay produces genuine sliding toward the most singular stratum, is untouched.

7 Reproducibility

All runs are deterministic on CPU with per-run manifests (SHA256 + git + config); raw episode files and aggregate JSON are committed alongside the code. Pre-registration commits precede every confirmatory run in the repository history.

- **Theorem constants and figures.** `06-notas/t2-prova/`: lemma verifications `passo0--6.py`, `prova_L1--L6.py`; figures via `figuras_draft.py`.
- **MLP mechanism (pilot).** `etapa4-mlp-populacao/pilotos-4a2/pressao/`; commit `b48b527`.
- **MLP confirmatory (4a.3).** `.../pressao/taxa/`: pre-registration `a8f8f8c`, run `5a961cb`. `python run_4a3.py --workers 4` then `python analise_4a3.py` (reads `results-4a3/` + `config_4a3.json`). Rate pilot `995a87c`. Staircase figure: `06-notas/paper/figuras_paper.py`.
- **Transformer (etapa-5).** `05-experimentos/etapa5-transformer/`: scoping `d4e4c35`, rate pilot `2dba27d`, confirmatory pre-registration `1b45498`.
  `python run_e5.py --workers 4` then `python analise_e5.py`.

- **Negatives.** `05-experimentos/NEGATIVOS.md` and the etapa-4a note record the invalidated instrument (commit `6411ac8`) and the rejected redesign (`a413b19`); the falsified schedule is `p5-relu-morte-v2/` (commit `190d94a`).

A The six lemmas

Lemma 1 (normal form)**.** *In the gauge $w = e^{\beta z}$, fast time $\tau = \beta t$: $dw/d\tau = cR(t)\, w^2/(1+w)\,(1+E_\beta)$, $|E_\beta| \le \log(1+w)/(\beta|R|)$ — the quadratic normal form of De Maesschalck–Schecter [**?**] with $\varepsilon = \beta^{-1}$. With $u = 1/w$ and $\Phi(u) = \log u + u$ one has the exact quadrature $d\Phi/dt = -c\beta R(t)(1+E_\beta)$.*

Lemma 2 (B-subsystem and feedback)**.** *Under (H4) the B-field is affine with Lipschitz constant $L_B = \lambda_{\max}(G_B)$ and R^{hyb} is affine in θ_B. Integral Grönwall consuming the bound of Lem.* **??** *gives, conditionally on the layer being maintained (supplied by Lem.* **??***), the refined rate $\sup \|\theta_B^\beta - \theta_B^{\mathrm{hyb}}\| = O(\log\beta/\beta^2)$ — one full order better than the naive $O(\beta^{-1})$: in the bulk $u = O(\beta|A|)$, so the coupling is $O(\beta^{-2})$ there and $O(\beta^{-1})$ only in the $O(\beta^{-1})$-wide corner windows. Verified: metric $\sup\cdot\beta^2/\log\beta$ stable in $[4.2, 5.7]$.*

Lemma 3 (entry corner)**.** *$\dot z^0(t_{\mathrm{hit}}^-) = cR^{\mathrm{hyb}}(t_{\mathrm{hit}}) < 0$ by (H1). The exact identity $(1-\sigma)/\sigma = e^{-s}$ collapses the crossing-time difference to an elementary integral: $t_{\mathrm{cross}}^\beta = t_{\mathrm{hit}} + \frac{1}{\beta v}(1 + O(\delta_z) + o(1))$ with $v = c|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|$, and the exponentially weighted approach integral is $\frac{3/2}{\beta v}(1+o(1))$. Verified: $K_{\mathrm{teo}} = 7.04$ vs $K_{\mathrm{emp}} = 7.13$ (1.2%); weight constant to $< 1\%$.*

Lemma 4 (sticking and release, prescribed R)**.** *For prescribed R satisfying (H1)–(H2), with $A(t) = \int_{t_{\mathrm{hit}}}^t R$: the exact additive identity $d\Phi/dt = -c\beta R + c\log(1+w)$ holds (no division by R; the interior zero of R forced by (H1)–(H2) is harmless — the layer is held by the* area*, not the sign of R), and on compact subintervals of $(t_{\mathrm{hit}}, t_{\mathrm{exit}})$, $\Phi(u(t)) = -c\beta A(t)(1+o(1))$, hence $u(t) = c\beta|A(t)|(1+o(1))$ (depth $z \sim -\beta^{-1}\log\beta$), and u returns to $O(1)$ at $t_{\mathrm{exit}} + O(\beta^{-1})$ — no $\log\beta$ factor. Sub/super solutions from monotonicity in R give explicit constants; a perturbation δ of R shifts t_{exit} by $\le c\,\delta\,(t_{\mathrm{exit}} - t_{\mathrm{hit}})/|A'(t_{\mathrm{exit}})| + O(\delta^2)$.*

Lemma 5 (coupled bootstrap)**.** *Continuation with strict margins: the property "θ_B is $2M_2\log\beta/\beta^2$-close to the hybrid and the margin stays strictly in the layer after an $O(\beta^{-1})$ entry collar" is nonempty (Lem.* **??***), open (strict inequalities + continuity) and closed (the assumption returns the better constant M_2 via Lem.* **??***, and Lem.* **??** *with $\delta = O(\log\beta/\beta^2)$ keeps the layer up to $t_{\mathrm{exit}} - C_\varepsilon\log\beta/\beta$), with $M_2 = e^{L_B T} C_f C_4/c$ independent of the hypothesis for $\beta \ge \beta_0$; with the empirical $C_4 = 10.6$, $M_2 = 440.8$; the leading term of the theoretical C_4 is $2/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})) = 29.6$. The entry corner cuts initial area $\approx R^{\mathrm{hyb}}(t_{\mathrm{hit}})K/\beta$, advancing the exit by $1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})\beta)$ — the arrival speed cancels. Hence the two-term closed-loop law $\beta(t_{\mathrm{release}}^\beta - t_{\mathrm{exit}}) = (-1+\mathrm{corr}(\beta))/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})) + o(1)$. Verified: the two-term law reproduces the entire coupled sweep at 1.9%/0.7%/0.2%/0.1% ($\beta = 3\cdot 10^2 \ldots 10^4$).*

Lemma 6 (exit corner; scalar reduction)**.** *Post-release the field is transversal outward ($cR^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$, (H2)); the mirrored corner identities of Lem.* **??** *apply with $v' = cR^{\mathrm{hyb}}(t_{\mathrm{exit}})$. Forward uniqueness of the restarted ReLU flow holds on the* activating branch*, selected by the limit itself (release at $t_{\mathrm{exit}} + O(\beta^{-1})$ is incompatible with the frozen Filippov branch); classical Grönwall glues on $[t_{\mathrm{release}}, T]$ at rate $O(\log\beta/\beta)$. Under (H3) the other data contribute $O(e^{-\beta\delta_1})$, the face-j field is rank one along $(x_j, 1)$, and the episode reduces to the scalar problem. Verified: gluing sup $= 1.6\cdot 10^{-3}/5.5\cdot 10^{-4}$ at $\beta = 10^3/3\cdot 10^3$.*

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